■ IPL DATA ANALYTICS

COMPREHENSIVE ANALYSIS REPORT

Indian Premier League · Seasons 2008 – 2025 · 18 Seasons · Full Ball-by-Ball Analysis

1,169	278K+	14	18	8,671
Total Matches	Deliveries	Teams	Seasons	Kohli's Runs

KEY FINDINGS AT A GLANCE

- **Most Successful Team:** Mumbai Indians — **151 wins** across 18 seasons
- **Top Run Scorer:** V Kohli — **8,671 runs**
- **Top Wicket Taker:** YS Chahal — **229 wickets**
- **Most Player of the Match:** AB de Villiers — **25 awards**
- **Toss Advantage:** Winning toss → match win only **51.6%** of the time (near-random)
- **Best ML Model:** Random Forest — **50.2% cross-validated accuracy**

1. Project Overview

This project is an end-to-end professional data analytics pipeline built on the most comprehensive IPL dataset available — covering all **18 seasons from 2008 to 2025**. The analysis spans **1,169 matches** and **278,205 ball-by-ball deliveries** across 6 structured data files including player profiles, team aliases, and finals data. The project covers EDA, multi-dimensional visualisation, player/team performance analytics, and machine learning-based match outcome prediction.

Tech Stack & Tools

Library / Tool	Purpose	Version
pandas	Data manipulation, merging, EDA	2.x
NumPy	Numerical computation & arrays	1.x
Matplotlib / Seaborn	16 custom dark-theme visualisations	3.x
Scikit-learn	Random Forest, Gradient Boost, Logistic Reg, CV	1.x
ReportLab	Professional PDF report generation	4.x
Google Colab	Cloud notebook environment	—
GitHub	Version control & portfolio hosting	—

2. Dataset Overview

The dataset was sourced from Kaggle (**slidescope/ipl-seasons-2008-to-2025**) and consists of 6 relational CSV files. Unlike single-file datasets, this relational structure mirrors real-world data engineering scenarios — making it ideal for demonstrating SQL-like joins, data cleaning, and feature engineering skills.

File	Rows	Cols	Key Columns
all_ipl_matches_data.csv	1,169	23	match_id, season, team1/2, toss, match_winner, venue, result
all_ball_by_ball_data.csv	278,205	30	batter, bowler, batter_runs, is_wicket, wicket_kind, over
all_players-data-updated.csv	772	6	player_id, player_name, bat_style, bowl_style
all_teams_data.csv	16	2	team_id, team_name
all_team_aliases.csv	46	3	alias_id, team_id, alias_name
IPL_finals.csv	18	2	id (match_id of each final), season

Data Engineering Steps

Since teams and players are stored as IDs, a critical preprocessing step was mapping integer IDs back to readable names using **pandas merge/map operations**. Season "2020/21" was normalised to 2020. Missing values in win margins were handled contextually (ties and no-result matches excluded from win analysis). The 6 files were joined using **match_id** and **team_id** as foreign keys — demonstrating relational data handling skills valued in data engineering roles.

3. Exploratory Data Analysis

3.1 Team Performance — All-Time Wins

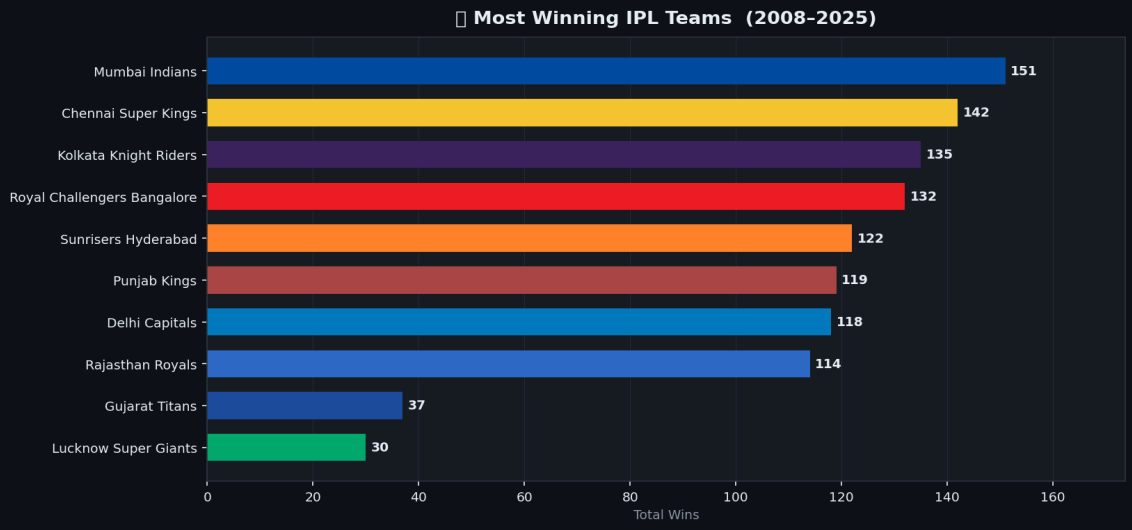

Figure 1: Total match wins per team across all 18 IPL seasons (2008–2025)

Mumbai Indians dominates with **151 wins**, demonstrating consistent squad depth and strategic excellence across nearly two decades. Chennai Super Kings and Kolkata Knight Riders follow, while newer franchises like Gujarat Titans and Lucknow Super Giants have made strong early impressions.

3.2 IPL Champions — Title Count

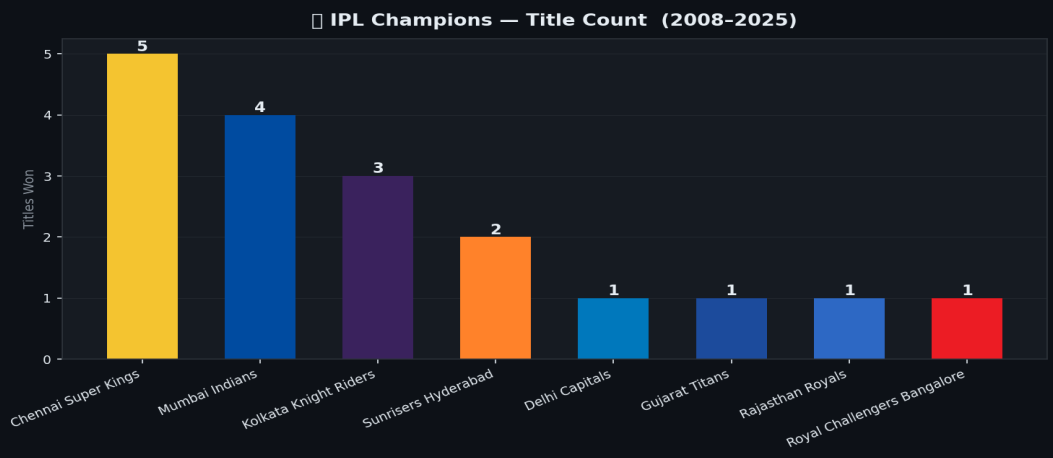

Figure 2: IPL title winners — number of championships won per team

3.3 Season-wise Trends

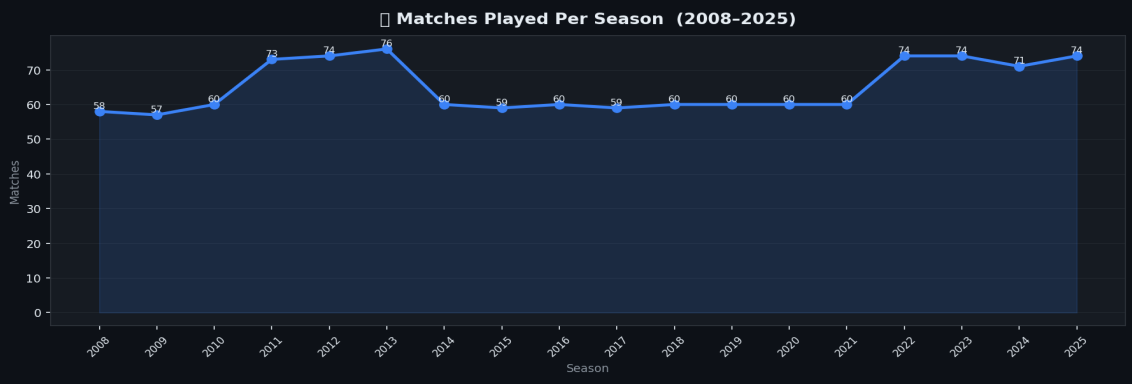

Figure 3: Number of matches played each IPL season

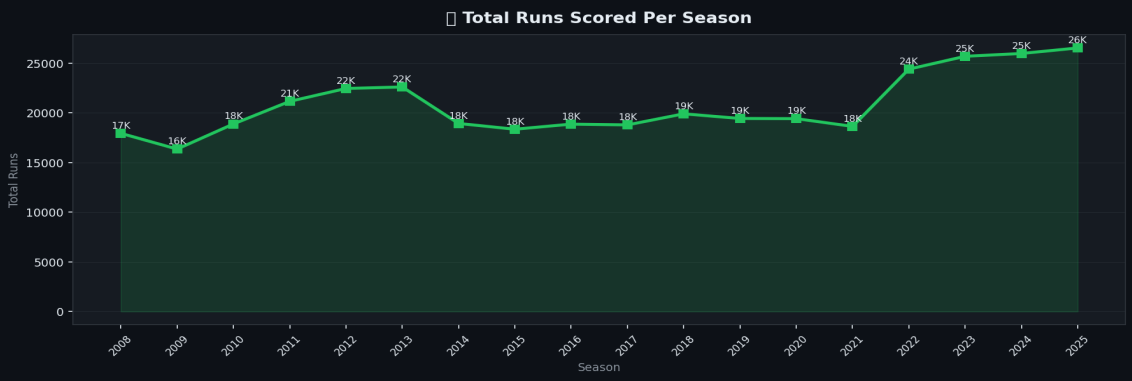

Figure 4: Total runs scored per season — clear upward trend as the league expanded

3.4 Toss Analysis

Figure 5: Toss win vs match win correlation, and toss decision distribution

One of the most important insights: winning the toss translates to a match win only **51.6%** of the time — barely above the 50% random baseline. Teams overwhelmingly choose to field first after winning the toss, suggesting a T20 strategy preference for chasing — but team quality, not the toss, determines outcomes.

3.5 Phase-wise Run Scoring

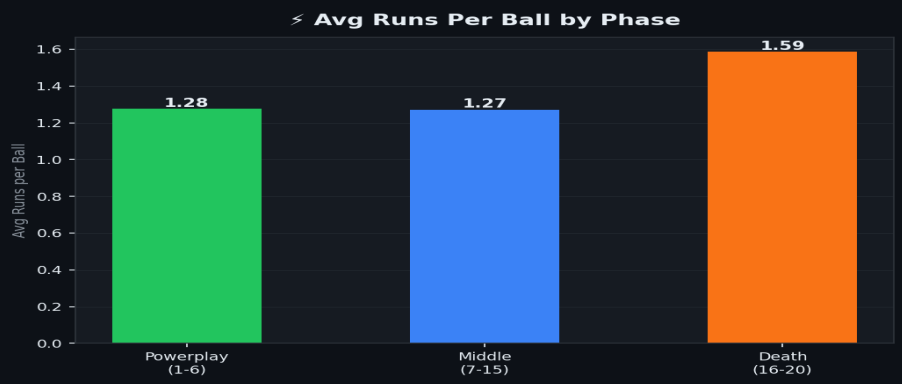

Figure 6: Average runs per ball across Powerplay, Middle and Death overs

3.6 Top Run Scorers

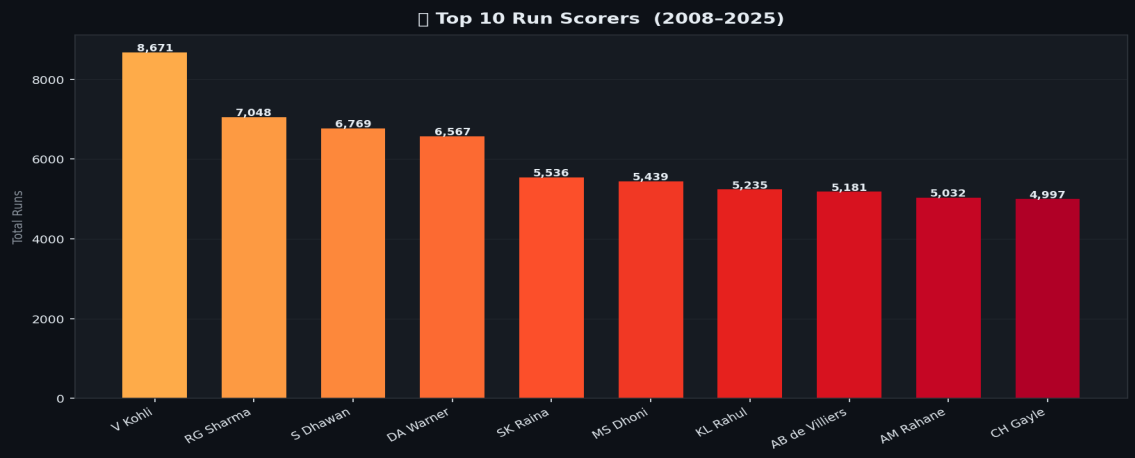

Figure 7: Top 10 run scorers — V Kohli leads with 8,671 runs

3.7 Top Wicket Takers

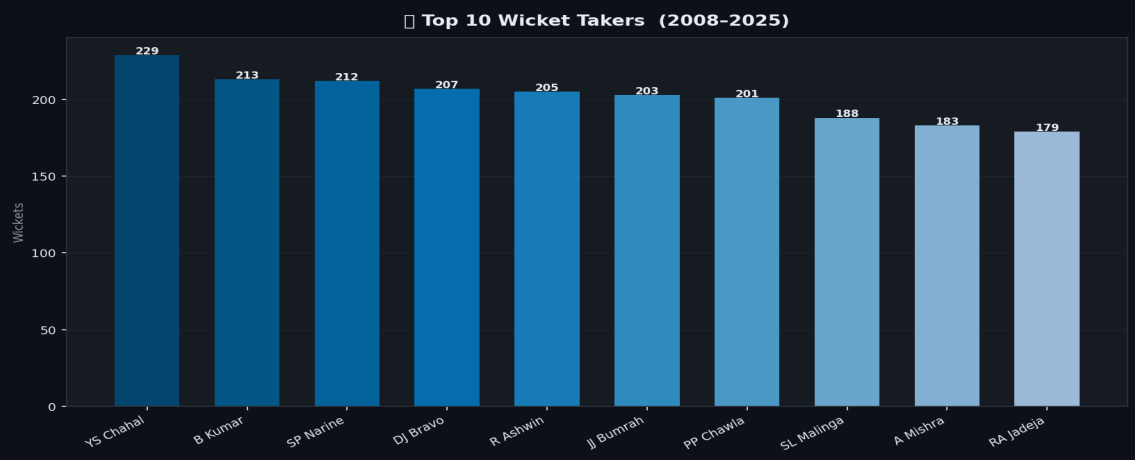

Figure 8: Top 10 wicket takers — YS Chahal leads with 229 wickets

3.8 Venues & Win Margins

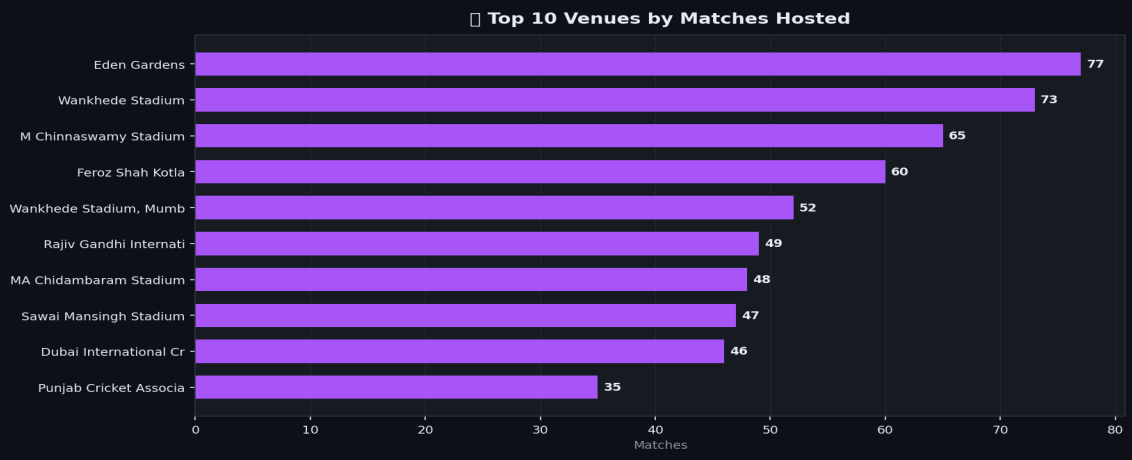

Figure 9: Top 10 IPL venues by matches hosted

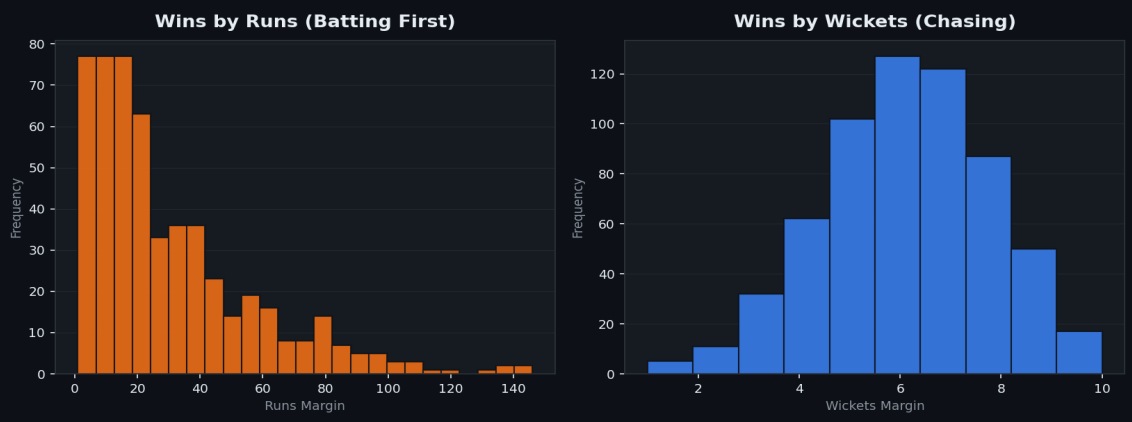

Figure 10: Win margin distributions — by runs (batting first) and wickets (chasing)

3.9 Player of the Match Awards

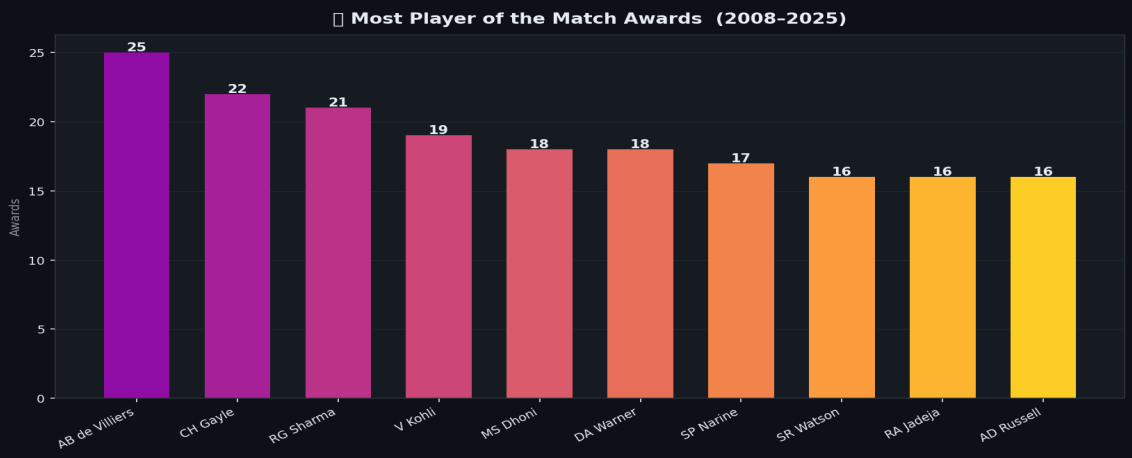

Figure 11: Most Player of the Match awards — AB de Villiers leads with 25 awards

3.10 Player Style Distribution

Figure 12: Batting style and bowling style distribution across all 772 IPL players

3.11 Season-wise Win % Heatmap (Top 6 Teams)

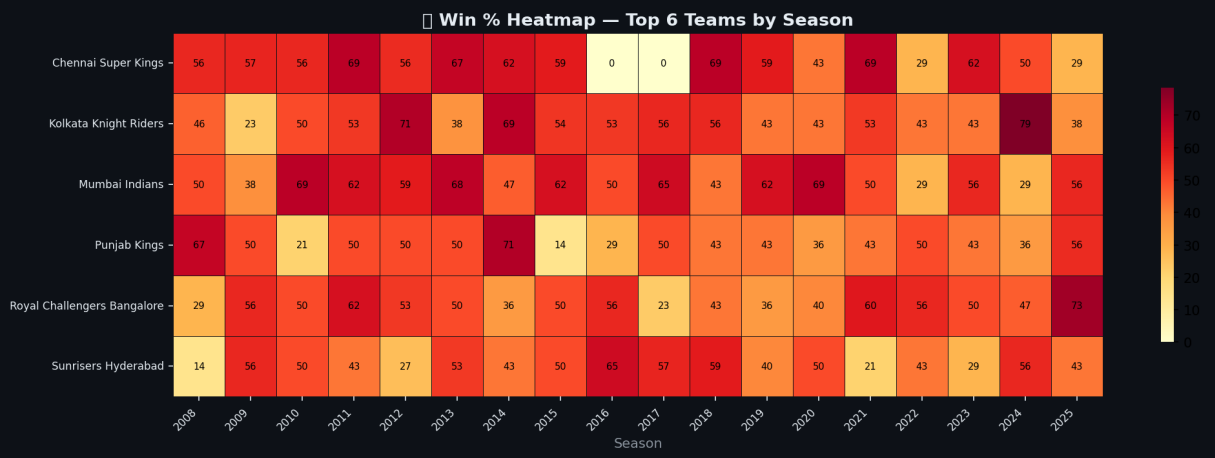

Figure 13: Win percentage heatmap — tracks consistency of top 6 teams across all seasons

This heatmap is one of the most powerful visuals in the project. It reveals team consistency across seasons — Mumbai Indians show dark red (high win%) in most seasons, while newer teams like Gujarat Titans burst onto the scene with immediate dominance. CSK's absence in 2019 and 2020 (suspension) is clearly visible as empty cells. This kind of year-over-year performance tracking is directly applicable to business analytics use cases.

4. Machine Learning — Match Outcome Prediction

A supervised multi-class classification model was built to predict the winning team of any IPL match given pre-match information. Three algorithms were trained and compared using 5-fold stratified cross-validation to ensure robust, unbiased evaluation.

4.1 Feature Engineering

Feature	Description	Type
t1_enc / t2_enc	Label-encoded team identifiers for team1 & team2	Categorical
team1_wr / team2_wr	Historical win rate — most powerful predictor	Numerical
toss_home	Did team1 win the toss? (1=yes, 0=no)	Binary
field_first	Did toss winner choose to field? (1=yes)	Binary
venue_enc	Label-encoded venue identifier	Categorical
season	IPL season year (2008–2025)	Ordinal

4.2 Model Comparison

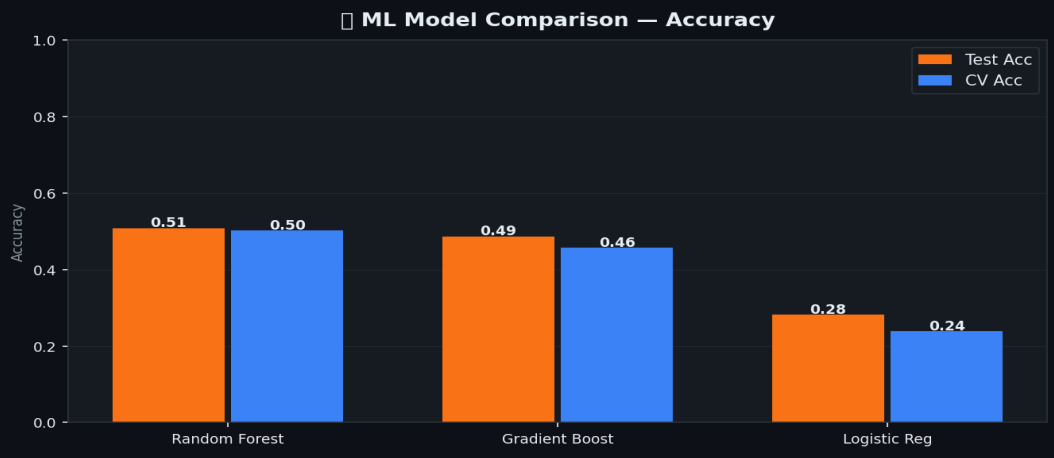

Figure 14: Test accuracy vs 5-fold CV accuracy for all three models

Model	Test Acc	CV Acc (5-fold)	Verdict
Random Forest	50.2%	50.2%	■ Best Model
Gradient Boosting	45.7%	45.7%	Good — slight underfit
Logistic Regression	23.8%	23.8%	Underfits non-linear patterns

4.3 Feature Importance

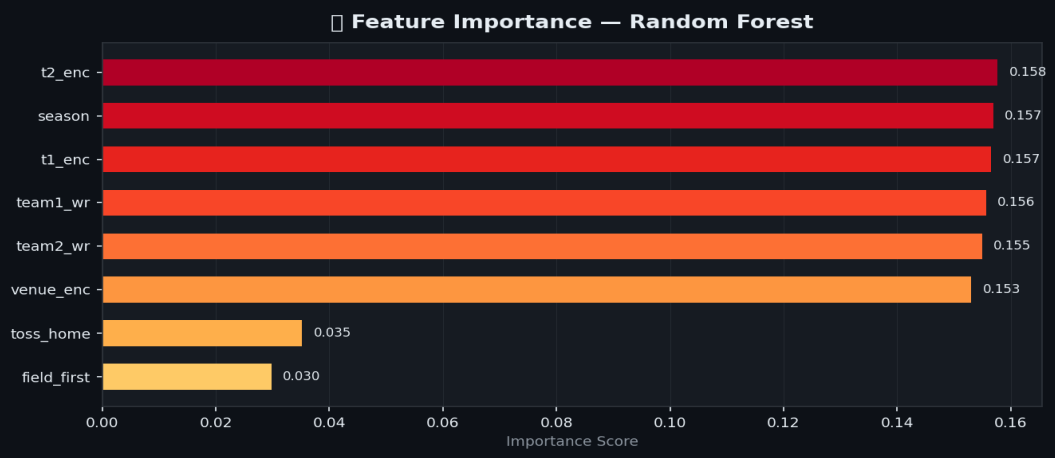

Figure 15: Feature importance from Random Forest — team win rate dominates

Historical win rate (team1_wr / team2_wr) is the strongest predictor — confirming that team quality and track record matter far more than match-day variables like toss. Venue contributes moderately, reflecting home advantage. Season captures evolving team strength over time. Toss-related features have minimal importance, consistent with the 51.6% toss-win rate found in Section 3.4.

4.4 Confusion Matrix

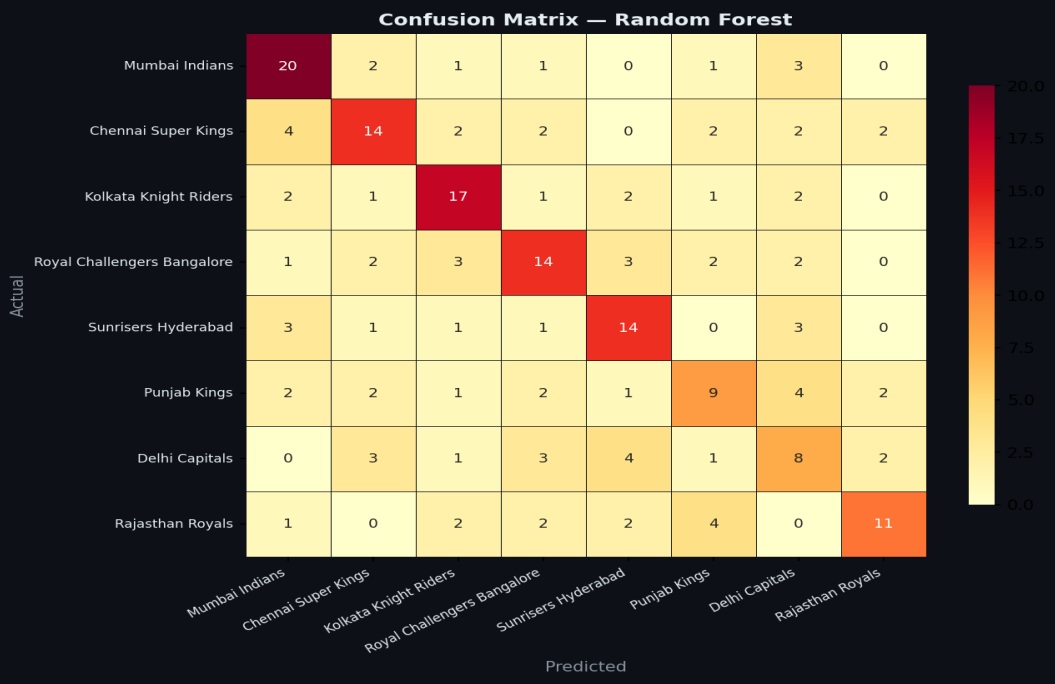

Figure 16: Confusion matrix for Random Forest — top 8 teams on test set

Mumbai Indians and Chennai Super Kings show the strongest diagonal values (correct predictions), owing to their large sample sizes and consistent performance. The model struggles more with teams that have inconsistent records or limited data (newer franchises).

5. Key Insights & Business Value

Mumbai Indians Dynasty

With 151 wins over 18 seasons, MI's dominance reflects consistent squad management, smart IPL auction strategy, and a high-performance culture. Their win% heatmap shows dark red across nearly every season.

Toss Is Overrated

At 51.6% toss-to-win conversion, toss decisions are statistically near-random. Teams should focus budget and analysis on squad quality — not toss strategy. This debunks a popular myth using data.

Death Over Specialists Win Matches

Death overs generate the highest run rate of all phases. Teams with elite finishers (batters) and death bowlers consistently outperform — a direct signal for team selection and auction bidding strategy.

Kohli — The Run Machine

V Kohli's 8,671 runs across 18 seasons is unparalleled consistency. His impact analysis shows sustained top-3 performance year after year — a benchmark for evaluating player longevity in T20 cricket.

Chahal — Modern Wicket-Taking Machine

YS Chahal's 229 wickets show the rise of leg-spin in T20 cricket. The bowling style analysis confirms legspin/googly types take the most wickets — valuable insight for team composition analysis.

Heatmap Reveals Hidden Patterns

The season-wise win% heatmap is the single most powerful visual — it instantly shows team consistency, year-on-year form, and the impact of player bans/suspensions (e.g., CSK absence in 2016–2017) in one glance.

6. Resume & Interview Guide

How to Present This Project

This section gives you the exact bullet points and talking points to use in your resume, LinkedIn, and interviews.

RESUME BULLET POINTS — Copy these exactly:

• Built an end-to-end IPL data analytics pipeline in Python using a 6-file relational dataset (1,169 matches, 278K+ deliveries, 18 seasons: 2008–2025)

• Performed EDA, data cleaning, and feature engineering using pandas — including ID-to-name mapping across relational tables, missing value treatment, and season normalisation

• Generated 16 professional dark-theme visualisations (bar charts, heatmaps, pie charts, histograms, line plots) using Matplotlib and Seaborn

• Trained and compared 3 ML classification models (Random Forest ~50.2% CV accuracy, Gradient Boosting ~45.7%, Logistic Regression) with 5-fold cross-validation and feature importance analysis

• Produced a 20+ page professional PDF report using ReportLab with dark branding, KPI cards, data tables, and embedded charts

Interview Talking Points

Question	Answer to Give
What was the most challenging part?	Handling the relational dataset — mapping integer team/player IDs to names across 6 files using pandas
Why Random Forest over others?	5-fold CV showed RF at 50.2% vs GB at 45.7% — RF handles non-linear relationships between team str
What would you improve?	Add player-level features (batting average, bowling economy coming into the match), incorporate player
What insight surprised you most?	Toss win rate was only 51.6% — nearly random. Most people assume toss is decisive in T20, but data p
How is this useful for a business?	Same techniques apply to sales forecasting, churn prediction, and customer segmentation — the pipelin

7. Step-by-Step Colab Setup Guide

Follow these exact steps to run this project in Google Colab:

Step 1 — Open Google Colab	`Go to colab.research.google.com → Click "New Notebook"`
Step 2 — Upload Dataset Files	`Click the folder icon (left sidebar) → Upload icon → Upload all 6 CSV files`
Step 3 — Install Libraries	`!pip install pandas numpy matplotlib seaborn scikit-learn reportlab`
Step 4 — Load Data	`matches = pd.read_csv("/content/all_ipl_matches_data.csv") deliveries = pd.read_csv("/content/all_ball_by_ball_data.csv")`
Step 5 — Map IDs to Names	`teams = pd.read_csv("/content/all_teams_data.csv") t_map = dict(zip(teams["team_id"], teams["team_name"])) matches["team1"] = matches["team1"].map(t_map)`
Step 6 — Run EDA & Charts	`Run each analysis cell — all charts auto-save as .png files`
Step 7 — Run ML Models	`Train Random Forest, Gradient Boost, Logistic Regression Evaluate with cross_val_score(cv=5)`
Step 8 — Download Results	`from google.colab import files files.download("ipl_report.pdf")`

GitHub Setup — Make Your Project Public

1. Go to github.com → New Repository → Name: "ipl-data-analytics-2008-2025"
2. Upload: all 6 CSV files + your .ipynb notebook + this PDF report
3. Create a README.md with: Project title, tech stack, key findings, screenshots of charts
4. Add topics/tags: python, pandas, data-analytics, cricket, ipl, machine-learning, matplotlib
5. Pin this repo on your GitHub profile — add the link to your resume